

BEYOND THE PALEOLITHIC PRESCRIPTION: INCORPORATING DIVERSITY AND FLEXIBILITY IN THE STUDY OF HUMAN DIET EVOLUTION

Evolutionary paradigms of human health and nutrition center on the evolutionary discordance or “mismatch” model whereby human bodies, reflecting adaptations established in the Paleolithic era, are ill-suited to modern industrialized diets resulting in rapidly increasing rates of chronic metabolic disease. Whereas this model remains useful, we argue that its utility in explaining the evolution of human dietary tendencies is limited. The assumption that human diets are mismatched to our evolved biology implies that they are instinctual or genetically determined and rooted in the Paleolithic. We review current research indicating that human eating habits are primarily learned through behavioral, social and physiological mechanisms starting in utero and extending throughout the life course. Those adaptations that appear to be strongly genetic likely reflect Neolithic, rather than Paleolithic, adaptations and are significantly influenced by human niche-constructing behavior. Incorporating a broader understanding of the evolved mechanisms by which humans learn and imprint eating habits and the reciprocal effects of those habits on physiology would provide useful tools for structuring more lasting nutrition interventions.